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RelaxVR: Cybersickness Reduction in Immersive Virtual Reality through Explainable AI and Large Language Models

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ABSTRACT

Virtual reality (VR) systems are susceptible to cybersickness, significantly hindering user immersion. Very recently, researchers introduced explainable artificial intelligence (XAI) enabled methods for detecting and explaining cybersickness features. Since XAI methods can identify the dominant features causing cybersickness, we argue that this knowledge can also guide the dynamic adoption of effective cybersickness reduction strategies, unlike state-of-the-art static approaches that rely on fixed methods. This paper introduces a new dataset *MazeSick* and *RelaxVR*, an interactive XAI-guided VR cybersickness reduction framework to predict, explain, and reduce cybersickness. Specifically, we propose an innovative XAI-guided cybersickness reduction engine, which selects the most appropriate reduction technique based on XAI-provided feature importance scores from a cybersickness reduction library of different reduction techniques. We also design an interactive dialogue system powered by large language models, enabling users to engage in the VR simulation via voice commands to understand the reasoning of cybersickness and select effective reduction strategies through natural language interaction. We deployed RelaxVR on a consumer-grade VR headset (e.g., HTC VIVE Pro) and validated it through a user study on cybersickness evaluation, where participants were immersed in a custom-built VR Maze simulation. Our results demonstrate that RelaxVR effectively reduces cybersickness with minimal impact on immersion, and 94% of the users agreed that RelaxVR is highly effective for reducing cybersickness and easier to use through the dialogue engine for selecting effective reduction strategies.

INDEX TERMS

 Cybersickness, Explainable AI, Virtual Reality, Large Language Model

I. INTRODUCTION

DESPITE the immense popularity of virtual reality (VR) in many application domains, such as education [1], military [2], health care [3], gaming [4], etc., cybersickness is still a major obstacle to their broader acceptability in our day-to-day life [5], [6]. Cybersickness may cause severe discomforts such as eye strain, nausea, disorientation, headaches, vertigo, and many more [7], [8] while using head-mounted displays, and these symptoms can last up to five hours after VR immersion [9]. Several methods exist in the literature on detecting cybersickness, including those that use pre/post-immersive questionnaires (e.g., simulator sickness questionnaire (SSQ), virtual reality sickness questionnaire (VRSQ), etc.) [6], [10] and those which are based on objective measurement based methods (e.g., heart rate (HR), eye-tracking,

head-tracking, etc.) [11]–[13]. While these questionnaire-based methods have been instrumental in identifying cybersickness symptoms, they possess inherent limitations. Notably, they are retrospective, capturing symptoms only after VR exposure, precluding real-time monitoring and intervention [12], [14]–[16]. At the same time, objective measurement methods allow for automated cybersickness measurement. Thus, recently, researchers have applied different machine learning (ML)/deep learning (DL)-based approaches that utilize objective measurement data (e.g., eye-tracking, head-tracking, etc.) to automatically detect cybersickness severity during VR immersion [14], [17], [18]. Unfortunately, ML/DL-based cybersickness detection models are often considered black boxes. Adding explainability to these models can provide several benefits, such as (i) helping in improving

the model's understanding and providing insight into why and how the model arrived at a specific decision and (ii) aiding designers to develop more effective cybersickness detection models by identifying the important features leading to cybersickness. Motivated by this, a new research direction has emerged recently, leveraging ML/DL models with explainable artificial intelligence (XAI) for improved cybersickness detection [19]–[21].

Limitations of existing approaches. Prior studies have applied various *static* cybersickness reduction techniques, such as field of view (FOV) manipulation, depth of field blurring, etc., to reduce discomfort by adding visual effects to virtual environments [22]–[32]. However, these methods have a few limitations: (i) State-of-the-art (SOTA) cybersickness reduction methods often require human intervention or blindly apply a *fixed/static* cybersickness reduction technique, such as FOV manipulation or depth of field blurring that are uniformly applied to all users. These "one-size-fits-all" approaches do not account for individual differences in cybersickness susceptibility, potentially disrupting user immersive experience and reducing overall effectiveness. This presents a significant challenge, as cybersickness is highly subjective [33]–[36] meaning that a reduction technique effective for one VR user may not be effective for another user [37]. Therefore, instead of blindly applying a specific reduction technique, it is required to understand the reasons behind the cybersickness model's decision by analyzing key features contributing to its prediction (e.g., causing cybersickness) through *explainability*, enabling adaptive and personalized cybersickness reduction, (ii) Prior works have explored ML/DL-based cybersickness detection and then apply reduction technique, but they often fail to provide real-time adaptability [32], [38]. Indeed, these works apply static reduction techniques without dynamically adjusting to users' physiological or behavioral changes during VR exposure. For instance, Islam et al. [32] introduced the Cybersense framework, which adaptively adjusts reduction techniques but does not account for the underlying causes of cybersickness and lacks real-time adaptability, potentially impacting the user's immersive experience. This limitation hinders timely interventions from preventing symptom escalation, underscoring the need for systems to detect and reduce cybersickness in real-time, and (iii) Prior works often lack mechanisms for users to interact with cybersickness detection models. The absence of user feedback or control over reduction strategies can limit the effectiveness of these systems and reduce user trust and comfort. Moreover, user interaction with an automated cybersickness detection model via a dialogue engine can significantly impact effective cybersickness reduction. For instance, a VR user uses a headset with automated cybersickness detection and reduction systems. The user is playing a game and wants to prevent cybersickness for himself. After playing for a minute, the user begins to feel cybersickness (e.g., dizziness). Then, the VR user interacts with the cybersickness detection model and listens to the explanation regarding the recommended reduction (XAI-guided reduc-

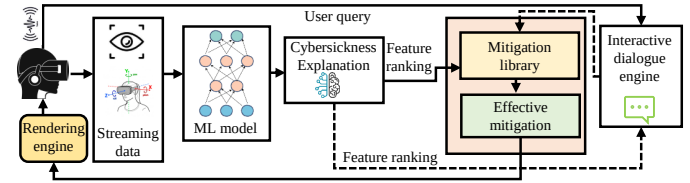


FIGURE 1: An overview of RelaxVR framework.

tion) technique. This explanation helps the user decide which reduction techniques to enable, allowing him to continue playing with reduced discomfort and a more enjoyable experience.

Key contributions. To address these research gaps, we propose a novel framework *explainable artificial intelligence guided interactive VR cybersickness reduction (RelaxVR)* to predict, explain, and reduce cybersickness in real-time by leveraging XAI and a large language model (LLM)-powered dialogue engine, as shown in Figure 1. Our contributions in this paper can be summarized as follows.

- We construct a new VR cybersickness dataset, *MazeSick*, which includes time-series VR integrated sensor data from multiple modalities, such as eye and head movement. Each sample is labeled by its corresponding cybersickness level collected from 60 participants through a user study using the VR Maze simulation. Our dataset is open source and thus can be used by other researchers in the community for further research¹.
- Next, we develop a Transformer and an ensemble ML model to demonstrate the MazeSick dataset's utility and deploy on a consumer-grade VR headset (HTC VIVE Pro) for real-time cybersickness detection from streaming eye-tracking and head-tracking data. Our results show that the deployed ensemble ML model in the VR headset classifies cybersickness with a high accuracy of 96%.
- We develop an effective method to select the appropriate cybersickness reduction technique from a cybersickness reduction library by mapping the predicted cybersickness explanation to the proper cybersickness reduction method. Our developed method can operate in two user-specified modes: 1) *Automated*—XAI-guided automatic cybersickness reduction without VR user intervention during immersion and 2) *Interactive*—XAI and large language models (LLMs)-based interactive dialogue system that enables VR users to interact with cybersickness detection models via voice commands to understand the reasoning behind cybersickness and how to choose effective cybersickness reduction techniques during immersion.

We validate the effectiveness of our proposed RelaxVR framework by conducting a user study using the same Maze simulation used for MazeSick dataset construction. Our experimental results show that RelaxVR significantly reduces cybersickness and provides a higher sense of presence with

¹<https://github.com/dependable-cps/RelaxVR>

marginal impacts on users' immersive experience (UIX) by dynamically choosing the proper reduction technique, unlike traditional methods, where the reduction methods are static.

II. RELATED WORKS

This section provides a brief discussion and review of the current understanding of cybersickness, highlighting several key aspects relevant to our work. SOTA works in cybersickness measurement can be broadly divided into three categories (i.e., subjective measures of cybersickness, objective measures of cybersickness, and ML/DL-based automatic cybersickness detection). This section discusses the following components in detail: subjective measures of cybersickness, objective measures of cybersickness, presence and cybersickness, ML/DL-based automatic cybersickness detection, and cybersickness reduction in VR. These components aim to provide a comprehensive understanding of the existing challenges and underpin the motivation for our research endeavor, highlighting the need for XAI and LLM in the development of effective cybersickness reduction methods.

A. SUBJECTIVE MEASURES OF CYBERSICKNESS

Cybersickness refers to the discomforts users experience while using virtual environments. Cybersickness may be explained by sensory conflict and postural instability theories [39]. According to these theories, symptoms of cyber sickness, such as eye strain, dizziness, headache, pallor, vertigo, sweating, disorientation, and vomiting, occur when the movement felt by the vestibular system in the brain is different from the screen seen with the eyes or when the posture is not maintained stably [18], [39]–[41]. To measure cybersickness, researchers have proposed several pre/post-questionnaire-based subjective measurements, such as the Simulator Sickness Questionnaire (SSQ) [10], [42], the VR sickness questionnaire (VRSQ) [6], [43], the Motion Sickness Susceptibility Questionnaire (MSSQ) [44], VR Neuroscience Questionnaire [45], etc.,. The SSQ comprises 16 questions divided into three categories (i.e., nausea, oculomotor, and disorientation) to assess the severity of each potential manifestation of cybersickness [42]. However, these post-immersive questionnaires cannot measure cybersickness during VR immersion and thus failed to provide a sufficient granular understanding of cybersickness severity during VR immersion. Motivated by this, authors in [16] proposed the Fast Motion Sickness Scale (FMS) Questionnaire to measure cybersickness severity during immersion. Indeed, recent work also uses a shortened version of the FMS Questionnaire to measure cybersickness during VR immersion using a Likert scale ranging from 0 to 10 (i.e., 0 means no cybersickness and 10 is the highest level of cybersickness) [14], [21], [33], [46], [47].

B. OBJECTIVE MEASURES OF CYBERSICKNESS

In addition to subjective measures of cybersickness, many researchers have also proposed objective measurements (e.g., heart rate (HR), eyeblink rate, galvanic skin response (GSR),

and electroencephalogram (EEG) signals) for detecting cybersickness [11], [17], [48], [49]. They found a significant correlation between cybersickness and physiological responses to VR stimuli, such as EEG, GSR, and HR [48]. For instance, authors [48] found that HR and EEG delta waves correlate positively with cybersickness, whereas EEG beta waves correlate negatively. Recently, built-in HMD sensors including eye and head tracking data (e.g., gaze ratio, pupil dilation, blinking, saccades, etc.,) have been used to understand the causes and severity of cybersickness as well as to develop methods to mitigate or prevent cybersickness [6], [13], [50]–[53]. For instance, authors [50] investigated cybersickness using eye-tracking measurements such as the center gaze ratio and scan path length. On the other hand, Reddy et al. [52] showed that cybersickness occurred when the pupil diameter was changed. While, Kourtesis et al. [6] measured cybersickness based on user eye-tracking data (e.g., pupil size) while the user was immersed in a VR simulation. Furthermore, authors in [54] reported that eye-tracking data (e.g., pupil diameter) reflexes decrease after VR immersion. In contrast, Nam et al. [50] used eye movement data and reported that cybersickness changes the gaze scan path and the gaze center ratio. Therefore, these objective sensor-based measurement data, specifically integrated eye and head tracking data, offer a complementary and scalable solution for cybersickness measurement by enabling continuous monitoring without interrupting the VR experience. These methods can measure early signs of discomfort using data such as eye and head movements; unlike subjective questionnaires, which rely on user recall and are limited to post-experience assessment, sensor-based objective measurement approaches allow for automated cybersickness measurement. Motivated by this, recent research focuses more on automatic cybersickness severity measurement from data readily available from the integrated sensors (e.g., eye-tracking, head-tracking) of consumer-level HMDs [13], [20], [33], [53]. Indeed, using integrated sensor data from consumer-level HMDs is more practical for predicting cybersickness in real-time since the users do not need to wear any other external sensors (e.g., HR, GSR, etc.,). Furthermore, eye-tracking and head-tracking data have often been used to understand the causes and severity of cybersickness and to develop methods to mitigate or prevent cybersickness [14], [20], [55].

C. PRESENCE AND CYBERSICKNESS

A significant line of research has explored the relationship between presence and cybersickness. Presence refers to the psychological state of “being there” in a virtual environment and is often thought to influence or be influenced by cybersickness. While some studies suggest that higher presence may reduce cybersickness by enhancing immersion and focus, others report conflicting findings, indicating a more complex, potentially bidirectional relationship [56], [57]. Presence is typically measured using subjective post-exposure questionnaires, such as presence questionnaire (PQ) [58], Igroup presence questionnaire (IPQ) [59], ITC sense of presence inven-

tory (ITC-SOPI), Slater Usuh steed presence questionnaire (SUS-PQ) [60], etc., For instance, The IPQ, developed by Schubert [59], is a 14-item tool assessing spatial presence, involvement, and the perceived realism of the virtual setting. For instance, the PQ developed by Witmer [58] consists of 32 items rated on a seven-point scale, designed to assess dimensions such as realism, interactivity, and interface quality. Another widely used questionnaire is the IPQ developed by Schubert [59], which includes 14 items to measure spatial presence, user involvement, and the perceived realism of a virtual environment. A detailed discussion of cybersickness and presence measurement questionnaires can be found in [61]. Thus, understanding the dynamic interaction between cybersickness and presence remains essential for designing more immersive, comfortable, and user-adaptive VR systems.

D. ML/DL-BASED AUTOMATIC CYBERSICKNESS DETECTION

Researchers have proposed different ML and DL-based methods to detect cybersickness severity automatically from a variety of subjective measurements data (SSQ), objective measurements data (bio-physiological signals), and integrated sensors data (eye-tracking) in HMD [13], [17], [18], [21], [33], [53], [67]–[71], as shown in Table 1. However, subjective questionnaire-based methods rely on user recall, are limited to post-experience assessment, and cannot detect cybersickness severity during VR immersion. These limitations prevent timely intervention and are not well-suited for adaptive VR experiences [32]. In contrast, sensor-based objective measurement approaches allow for automated cybersickness measurement. Thus, previous ML and DL-based work measured cybersickness with objective measurement-based bio-physiological signals (e.g., HR, GSR, EEG, etc.) [19], [21], [72], [73]. However, collecting bio-physiological data requires external sensors such as HR, GSR, etc., which are hard to deploy as standalone cybersickness predictors in current consumer-level HMDs (e.g., HTC Vive, Meta Quest Pro, etc.) [72], [74]. Furthermore, users are often instructed to limit their movements during data collection to avoid noisy data [72], [74] and limit VR locomotion and 3D-object manipulation during the immersion. Motivated by this, researchers have recently measured cybersickness severity from data readily available from the integrated sensors (e.g., eye-tracking, head-tracking) of consumer-level HMDs. Indeed, using integrated sensor data from consumer-level HMDs is more practical for predicting cybersickness in real time since the users do not need to wear any other external sensors. For instance, authors in [20] proposed a cybersickness severity measurement method using different DL models utilizing built-in HMD sensors, including eye and head tracking data. While Jeong et al. [13] applied spatial and temporal Transformer encoder models (MS-STTN) to predict cybersickness from the multimodal sensor data (e.g., eye, head, and EDA). However, these ML/DL-based cybersickness detection models are perceived as black-box models lacking explainability. To provide an explanation of cybersickness detection,

a new research direction has emerged recently, leveraging ML/DL models with XAI for cybersickness detection [19]–[21]. For instance, authors in [20], [21] used DL-based methods, namely LSTM, GRU, and MLP, and ensemble-learning-based super-learner models for cybersickness severity detection and then applied a post-hoc explanation-based method, namely, SHAP, LIME, MSA, and PDP to explain the detected cybersickness from user bio-physiological, subjective, and integrated sensors measurements data. Moreover, they reduced the feature space based on their explanation for retraining the cybersickness detection models and revealed that normal eye origin, gaze origin, pupil diameter, HR, GSR, etc., are the most influential features for causing cybersickness. Despite the great prospect of XAI in explaining cybersickness and reducing the model size, these methods have not yet been explored for how we can build effective cybersickness reduction strategies through these explanations. For instance, authors in [21] deployed a super-learner model in an embedded device for cybersickness detection using XAI-guided feature reduction techniques. Unfortunately, these works primarily focus on the prediction of cybersickness without demonstrating how XAI-driven insights can guide effective reduction. They also lack a comprehensive system for real-time cybersickness detection and reduction in real-world deployment and do not include user studies to evaluate the VR experience.

E. CYBERSICKNESS MITIGATION

Various visual techniques have been explored to reduce cybersickness severity in immersive VR experiences [14], [22]–[24], [26], [28]–[30], [38], [75]–[77]. Researchers used dynamic field of view (DFOV)-based methods for restricting the FOV to reduce discomfort [77]–[81]. For instance, Fernandes et al. [79] proposed a DFOV adjustment technique to automatically adjust FOV depending on the linear and angular speed of the virtual scene. On the other hand, Budhiraja et al. [82] introduced the dynamic Gaussian blur technique to reduce the amount of optical flow available to the user to minimize the cybersickness. They observed that eye gaze behaviors differ during immersion for both foveated and fixed FOV restrictors, for which a small static central window was associated with slower increases in cybersickness. In contrast, Carnegie et al. [76] deployed a real-time dynamic depth-of-field (DOF) technique that maintains the screen's center in focus and reports that DOF blurring reduces the cybersickness. Unfortunately, their proposed blur effect was not always effective in lowering cybersickness. Similarly, Ang et al. [31] developed GingerVR, a Unity-based software framework for cybersickness reduction, adaptable to any Unity-developed VR simulation. More recently, adaptive VR approaches have emerged, aiming to detect cybersickness severity during immersion and dynamically enhance the experience without user intervention [32], [38]. For instance, Kim et al. [38] proposed an adaptive VR-based approach for cybersickness detection and reduction. Still, it relied on binary classifiers and applied a specific cybersickness reduction approach (i.e., reduce FOV), limiting its ability to scale reduction techniques

TABLE 1: Comparison of our proposed *RelaxVR* method with state-of-the-art related work in ML/DL-based model in cybersickness prediction (CS Pred.), where VR exposure indicates whether participants actively navigate the environment or passively watch VR content. Sensor refers to modality (E: eye movement, H: head movement, BP: bio-physiological-based external sensor signals), along with locomotion (Loc.) availability, VR exposure time, and number of participants (Part.).

Prior works	Year	VRexposure	Exposure Time	Sensor	Loc.	Part.	Model	CS Pred.	Performance
Jin et al. [62]	2018	Active	3 Minutes	H	No	24	LSTM	SSQ	0.86 (R)
Lee et al. [63]	2019	Passive	2 Minutes	BP	No	20	ConvLSTM	SSQ	7.34 (RMSE)
Kim et al. [12]	2019	Passive	0.5 Minutes	BP	No	202	CNN-RNN	CSL (5 classes)	0.89 (Acc)
Jeong et al. [64]	2019	Passive	1 to 5 Minutes	BP	No	25	CNN/DNN	CSL (2 classes)	0.98 (Acc)
Islam et al. [15]	2020	Passive	12 minutes	BP	No	23	CNN-LSTM	CSL (3 classes)	0.98 (Acc)
Islam et al. [14]	2021	Active	7 minutes	E/H/BP	No	26	CNN-LSTM	CSL (3 classes)	0.87 (Acc)
Wang et al. [65]	2022	Passive	7 minutes	E/H/BP	No	20	LSTM	SSQ	0.83 (Acc)
Qu et al. [66]	2022	Active	2-3 minutes	BP	No	9	LSTM-Attention	CSL (2 classes)	0.96 (Acc)
Kundu et. al [20]	2023	Active	7 minutes	E/H/BP	No	30	LSTM/GRU/MLP	CSL (4 classes)	0.94/0.93/0.89 (Acc)
Jeong et al. [51]	2023	Passive	3-4 minutes	E/H/BP	No	27	CNN & LSTM Attention	CSL (4 classes)	0.83 (Acc)
Kundu et. al [21]	2023	Active	7 minutes	E/H/BP	No	30	Super Learner	CSL (4 classes)	0.96 (Acc)
Li et. al [67]	2023	Active	4 minutes	E/H/BP	No	53	LSTM	CSL (4 classes)	0.90 (Acc)
Islam et. al [33]	2024	Active	7 minutes	E/H/BP	No	30	DeepTCN	CSL (0-10)	0.82 (RMSE)
Wen et al. [68]	2024	Passive	10 minutes	E/H/BP	Mixed	500	Self-supervised learning	CSL (3 classes)	0.79 (Acc)
Jeong et al. [13]	2024	Passive	12 minutes	E/H/BP	No	45	Transformer	CSL (0-10)	0.25 (RMSE)
Kundu et. al [53]	2024	Active	7 minutes	E/H/BP	No	30	CNN/LSTM/GRU/MLP	CSL (4 classes)	0.80/0.94/0.92/0.86 (Acc)
Proposed RelaxVR	2025	Active	12 minutes	E/H	Yes	60	Transformer/Ensemble-ML	CSL (0-10)	0.94/0.96 (Acc) & 0.16/0.12 (RMSE)

based on the intensity of cybersickness. Furthermore, authors in [32] introduced the Cybersense framework, which adaptively adjusts reduction techniques but does not account for the underlying causes of cybersickness, potentially impacting user immersive experience. Indeed, they used external sensor measurement data, which is not practical for a realistic use case of VR headsets (e.g., HTC VIVE PRO). Furthermore, prior VR frameworks often required humans to guide the user through the cybersickness mitigation phase. Incorporating a task-oriented dialog agent within such a framework has the potential to reduce subjectivity and human intervention while increasing UX. For instance, authors, in [83], developed an avatar and task-oriented dialog agent to monitor cybersickness during a VR cognitive exercise and provide personalized information and advice on its relief. However, these methods still require a high level of expertise because users must know how the conversation will help reduce cybersickness since they did not adopt any explicit reduction technique. However, we provide rich conversations with RelaxVR by incorporating LLM-based language understanding and explainability of the cybersickness model. Moreover, we develop a cybersickness reduction engine that triggers a reduction technique based on XAI-based feature importance scores to select the most suitable technique from a library of cybersickness reduction methods. To the best of our knowledge, XAI and LLM-based techniques for detecting and reducing cybersickness have not been explored yet, which motivates our work in the paper.

III. MAZESICK DATASET DEVELOPMENT

In this work, through a user study, we first develop a multimodal integrated sensor dataset for cybersickness in the VR (MazeSick) dataset to facilitate the training of the cybersickness prediction models. We prepared a virtual maze environment for data collection in which the participants were navigated with a smooth locomotion technique (e.g., using the controller) and performed coin collection tasks. It is worth mentioning that only a few open-source datasets containing data from the integrated sensors (e.g., eye-tracking and head-

tracking) exist. However, these datasets also integrate external physiological sensor measurements (e.g., heart rate (HR), electrodermal activity (EDA), etc.), as discussed in Table 1. Indeed, while using these external sensors, users are often instructed to limit their movements [14], [15], [72], [74], and these sensors also limit VR locomotion and 3D-object manipulation during the immersion. Furthermore, these external sensors often require tethering and affixing sensors to the users' hands [14] and are also prone to noise [72], [74], which is not conducive to room-scale VR experiences. Moreover, VR data collected in one environment (e.g., VR roller coaster) often lacks generalizability to other environments (e.g., VR maze) for accurate ML and DL-based cybersickness prediction. For instance, a VR roller coaster simulation involves rapid accelerations, sharp turns, and intense motion cues, primarily triggering vestibular conflicts, while a maze simulation emphasizes spatial navigation and cognitive load. These differing sensory stimuli lead to distinct cybersickness triggers, making models trained in one environment less effective in others. Similarly, variations in visual complexity, motion patterns, and user interactions across environments further reduce the generalizability of these ML/DL models across distinct VR experiences, thus highlighting the need for diverse datasets to ensure robust and adaptable prediction models. The following subsection details the user study of the data collection.

A. PARTICIPANTS, APPARATUS AND VIRTUAL ENVIRONMENT

This user study involves 60 volunteer participants who received no reward (39 males and 21 females) aged from 18 to 38 years (mean age 23.6, standard deviation (SD) 3.21). The participants were Asian, Black, and Hispanic. Thirty four of the sixty participants said they had previously used VR equipment, whereas the rest of them said they had never done so. No individuals reported vestibular dysfunction or the usage of anti-motion sickness medication.

The virtual environment was a Dungeon-style maze created



FIGURE 2: First-person point of view of the maze simulation for *MazeSick* dataset collection user study.

using Unity and OpenVR, which contained brick walls, brick platforms, and an open roof with an orange sky, as shown in Figure 2. The maze was structured as a 15×14 grid, where a trophy is randomly spawned. The goal is to reach the trophy, which will transport the user to the next randomly generated maze. Each maze is designed so that the user must explore at least half of it to ensure a consistent level of challenge across levels. The maze occupies a total area of 150 meters by 140 meters in Unity (*Vector3(150, 1, 140)*). Given this size, each platform within the grid measures 10 meters by 10 meters in real-life space, as Unity's default scale assigns 1 unit to 1 meter. It is a single-player experience where the objective is to escape the maze. They provide options for both room-scale and smooth locomotion. Room-scale movement is supported by direct tracking on the HTC Vive headset, enabling users to walk through the maze physically. However, due to the maze's size, users may also need to rely on teleportation or smooth locomotion, which can be controlled using the HTC Vive's trackpads. The game places players in a dark, confined space where they must navigate their way through the maze. During testing, users were tasked with reaching the end of the maze under various scenarios.

Our work used an HTC-Vive Pro Eye headset to render the VR maze simulation, which offers a display resolution of 2880×1600 pixels per eye and a refresh rate of 90Hz. The headset provided a wide field of view, spanning 110 degrees. To ensure a seamless VR experience, we configured the HMD with the Vive wireless adapter. The audio was delivered through the integrated Vive headphones. We use the HTC SRanipal SDK and Tobii HTC Vive Devkit [84] to capture eye and head-tracking data. The VR maze simulation was rendered using a computer equipped with an Intel Core i9 CPU with 128 GB of memory and an Nvidia GeForce RTX 3090 GPU. The virtual environment was built using the Unity 2021.2.2f1 game engine software and OpenVR software development kit (SDK).

B. DATA COLLECTION PROCEDURE

Our user study was approved by the Institutional Review Board (IRB) at the authors' university. During the user study, the participants read the informed consent form and filled

TABLE 2: List of the features in the *MazeSick* dataset.

Data type	Features
Eye-tracking	Left and right pupil diameter (mm)
	Left and right normalized gaze direction/origin (x, y, z)
	Left and right pupil position (x, y, z)
	Convergence distance (mm) % of eyes open
Head-tracking	Head Quaternion Rotation (x, y, z, w)
	Head Euler (x, y, z)

out the IRB-approved demographic information, such as age, gender, initial SSQ, disability questions, and whether they had prior immersive VR experience. Then, we gave them an overview of the activity during the study. They were also told that they could end the study at any moment if they became too uncomfortable. Afterward, they entered the VR simulation study and fitted with the HTC Vive Pro Eye headset and controllers; they were brought into a tutorial that explained the tasks throughout the maze.

Before starting the experiment for data collection, we briefly introduced the procedures and informed participants about the requirements. The participants went through a brief 2-minute virtual tutorial or training explaining the coin collection task they had to complete. They were then instructed to use the controller to complete the maze navigation with the learned task. Furthermore, We emphasized verbally reporting their discomfort on an FMS score of 0 to 10 whenever a pre-recorded voice prompt was played at each 30 seconds interval during the VR simulations. We chose to use an FMS scale from 0 to 10 to measure participants' cybersickness level in real-time and to be comparable to previous research in ML/DL-based cybersickness detection [13], [14]. A rating of 0 means no significant change in comfort level compared to the resting baseline (no sickness), and 10 means significant discomfort compared to resting conditions (severe sickness). The participants navigated the maze using the smooth locomotion technique. We used this technique in the VR maze navigation task to allow participants to navigate the virtual environment without the physical space limitations required for real walking. Smooth locomotion offers a seamless way to explore large virtual areas while minimizing the risk of collisions and maintaining a consistent user experience, making it a practical choice for studies conducted in controlled settings [85], [86]. It is important to note that the eye and head positions were calibrated before starting the maze simulation. Upon satisfactory calibration, participants were immersed in a VR trial session.

After completing the trial/tutorial session, the VR maze simulation was played and lasted 12 minutes long unless the participants decided to quit earlier. After completing the VR simulation, participants were asked to complete a post-VR SSQ questionnaire [39] and then rest. Furthermore, before participants left the room, we reminded them that they could reach out to us if they felt severe discomfort or experienced any negative impact from the study. So far, no concerns from any participants have been reported.

C. DATA PRE-PROCESSING

We collected eye and head-tracking data during the VR immersion, as shown in Table 2. We also collected FMSQ responses alongside these objective measures during the VR maze simulation. After collecting the data, all personal identifiers were removed to ensure privacy. Subsequently, we applied standard data pre-processing techniques, such as outlier detection, de-noising, normalization, etc., to prepare the dataset for subsequent analysis. At first, we performed outlier detection and removed all the outliers that were three standard deviations away from the mean value by applying the z-score analysis method [87]. We then utilized exponential smoothing [88] to process the collected time series data using the following formula: $\hat{y}_{t+1} = \alpha \cdot y_t + (1 - \alpha) \cdot \hat{y}_t$. Here, $\hat{y}_{t+1}$ is the forecasted value at time $t+1$, α is the smoothing parameter, a value between 0 and 1, y_t is the actual value at time t , and $\hat{y}_t$ is the forecasted value at time t . We set the α value to 0.8 through a trial and error approach, effectively reducing the noise optimally without deviating too much from the original signal. Finally, to ensure comparability and consistency between different measurement data (e.g., eye-tracking, head-tracking), we normalized [89] them using the following formula: $X_{\text{normalized}} = \frac{X - \mu}{\sigma}$. Here, $X_{\text{normalized}}$ represents the normalized sample, μ is the mean of the sample, and σ is the standard deviation of the sample. It is important to note that all the samples are time-synchronized, thus preserving the time dimension of the dataset.

1) Ground Truth Construction and Categorization

To establish the ground truth, we utilized FMS scores recorded at 30-second intervals throughout the VR maze simulation. While SSQ scores were collected after the simulation, SSQ is a post-immersive questionnaire that does not measure cybersickness in real time (during VR immersion). Thus, we used the FMS scores to construct the ground truth in this study. To support cybersickness classification tasks, we performed a distribution analysis of the gathered FMS scores from all participants, similar to prior research [11], [14], [15]. More specifically, we categorize the target values (e.g., FMS) into three categories, as shown in Equation III-C1.

$$CS_t = \begin{cases} \text{Low,} & \text{if } 0 \leq FMS_t \leq Q_1 \\ \text{Medium,} & \text{if } Q_1 < FMS_t \leq Q_2 \\ \text{High,} & \text{if } FMS_t > Q_2 \end{cases} \quad (1)$$

In Equation III-C1, FMS_t denotes the cybersickness severity (CS_t) type. The quantiles were defined as follows: $Q_1 = 1$ and $Q_2 = 3$. This deliberate choice of $Q_1 = 1$ and $Q_2 = 3$ was made to effectively classify cybersickness severity while ensuring dataset balance.

IV. CYBERSICKNESS PREDICTION MODEL DEVELOPMENT, TRAINING, AND DEPLOYMENT

It is crucial to predict cybersickness accurately for effective cybersickness reduction. Therefore, we develop one ensemble model using SOTA ML models and a DL-based Transformer model to build an accurate cybersickness prediction

model. The architecture of cybersickness prediction models is discussed in the following subsections.

A. ENSEMBLE ML MODEL

At first, we develop an ensemble model using four ML models, such as random forest classifier (RFC) [90], XG boost classifier (XGBC) [91], extra trees classifier (ETC) [92], and gradient boosting classifier (GBC) [93] as base learners and a logistic regression classifier [94] as a meta learner. It is important to note that the proposed ensemble ML method is based on two steps: (1) building base learners by stacking multiple ML models into an ensemble and (2) fitting the base learners with a meta learner. Next, we train the ensemble ML model from the historical data, i.e., using the VRWalking data [47]. Finally, the ensemble ML model is trained based on the selected hyperparameters from Scikit-learn [95] using the Grid search CV method [96], in which every learner is taken with individual hyperparameter values. For instance, the no. of estimators for ET, XGB, and RF classifiers are 50, 40, and 15 and the learning rate for GB and XGB classifiers are 0.05 and 0.03, respectively.

B. TRANSFORMER MODEL

A Transformer is a sequence-to-sequence architecture consisting of an encoder and a decoder [97]. The encoder takes the input sequence and maps it into a higher-dimensional vector, which is then fed into the decoder to generate an output sequence. In our work, the Transformer network consists of three sub-networks: (1) an embedding network, also known as a positional encoding (PE), that changes the dimension of input data and reflects positional information, (2) the encoder that trains the importance of data characteristics of time-series data through multi-head self-attention, and (3) the output layer that predicts the likelihood of cybersickness severity in the future through an activation function. At first, PE assigns positional information to each feature using sine-cosine positional coding to enable the cybersickness prediction model to understand the sequential information in the input features accurately. Then, it passes to the transformer encoder block, which consists of multi-head self-attention, add and normalization, and the feed-forward layer. In this work, we use a 64-dimensional attention vector, which goes through an attention process in the Transformer encoder. After that, we use the add and normalization layer, which uses the residual connection for each sub-layer to mitigate the problem of vanishing gradients when training networks. Lastly, in the feed-forward layer, the MLP network is applied to capture the latent feature, which comprises 256 dimensions in our case. We use the Adam optimizer to optimize our DL-based Transformer model with epochs of 200 and a batch size of 256. We use a learning rate of 0.001 to train this Transformer DL model. We also deploy an early-stopping strategy with a patience value of 30 while training the cybersickness Transformer model to prevent it from overfitting.

Furthermore, we use a 10 fold cross-validation technique to train and validate the performance of the ensemble ML and

Transformer models in which the dataset was partitioned into k groups (i.e., in our case, $k = 10$). Only one partition out of k is utilized to validate the model, while the remaining partitions are used for training. The method is repeated k times, each time picking a new test partition and the remaining $(k - 1)$ partition as a training dataset to eliminate bias. After training and validation, the cybersickness prediction model is converted to ONNX format [98] and deployed in VR headsets.

V. PROPOSED RELAXVR FRAMEWORK

This section presents the details of our proposed *RelaxVR* framework for real-time cybersickness detection and reduction, comprising two essential parts: (1) a local VR environment and (2) a cloud-based environment. The local VR environment hosts the VR simulation, an on-device cybersickness prediction model, and an XAI-guided cybersickness reduction engine. In contrast, the cloud-based environment features a dialogue engine that facilitates interaction with the cybersickness detection model, supported by an explanation engine and LLM-guided natural language conversations. Furthermore, our proposed *RelaxVR* has two user specified modes that a user can choose before starting a VR session: 1) Automated mode, where the VR user receives the XAI-guided automatic cybersickness reduction after flagged the cybersickness severity level (e.g., low, medium, and high) without user intervention during immersion and 2) Interactive mode in which a VR user may be involved in a task where he may or may not wish to apply cybersickness reduction. Thus, when cybersickness is detected, he is alerted first. The user can interact with the *RelaxVR* system via voice command to understand the reasoning behind the severity of the flagged cybersickness and find suitable reduction through the inherent use of the XAI and LLM-guided dialogue engine in the *RelaxVR* framework, or the user can ask *RelaxVR* his preferred cybersickness reduction to take place during the VR simulation. The details of the overall *RelaxVR* framework are described in the following sections.

A. LOCAL VR ENVIRONMENT

The proposed environment is developed as a Unity project compatible with VR headsets, such as the HTC VIVE Pro, equipped with eye and head-tracking capabilities. We employ the Tobii API [84] to stream real-time eye and head-tracking data, which is used to predict the severity of cybersickness through a pre-trained ML/DL model deployed on the headset. Upon predicting the severity level of cybersickness, the model forwards this prediction along with the eye and head tracking data to the explanation engine, which generates the feature importance scores necessary for XAI-guided cybersickness reduction. The details of the local VR environment are as follows.

1) Inference Phase

During the inference phase, the pre-trained cybersickness model (e.g., ensemble ML or Transformer) predicts the sever-

ity of the cybersickness (low, medium, and high) using real-time streaming eye-tracking and head-tracking data from the VR simulation. The cybersickness model predictions are then forwarded to the cybersickness reduction engine, which notifies the user of predicted cybersickness along with an explanation and provides guidance on selecting an appropriate reduction technique. Suppose the user agrees with the result of the cybersickness detection and wants to apply the reduction technique. Then, the prediction decision is forwarded to the explanation engine to apply the XAI knowledge-based recommended reduction. Conversely, if the user disagrees with the prediction, no reduction is applied within the rendering engine. The reason is that cybersickness is highly subjective and task-dependent [99]. For instance, experienced VR users often can tolerate medium levels of discomfort compared to novice VR users. Furthermore, Our architecture employs a stream learning approach called online learning or Incremental learning [19]. Unlike batch learning, Stream learning trains the model incrementally on a continuous data stream, allowing the model to learn sequentially from individual observations or small groups of observations. This approach is advantageous for real-time applications with rapid changes and limited computing resources. We utilized the River Library [20] for dynamic data streams and continual learning. We defined a pipeline that includes a scaler transformer and a logistic regression model with stochastic gradient descent optimization with a learning rate of 0.01. This incremental learning update is fast and computationally efficient.

2) Cybersickness Reduction Phase

Upon receiving the cybersickness explanation result as a feature ranking from the cloud server, it is forwarded to the cybersickness reduction engine via an HTTP request. The cybersickness reduction engine then sends the ranked features to the reduction library, where an appropriate reduction technique is identified by analyzing the correlation between features and available reduction strategies. Finally, the recommended cybersickness reduction technique is forwarded to the rendering engine to update the frame. The details of these steps are provided below.

Algorithm 1: XAI-guided cybersickness reduction

Inputs : List of cybersickness reduction technique CS_{reduc} ,
Feature importance scores F_s

Outputs : $Pr_{S_m} = \{ \}$

- 1: **for each** ranked feature $F_i \in F_s$ **do**
- 2: $Reduc_{rank} = CSReductionMatch(CS_{reduc}, F_i)$
- 3: $Pr_{S_m} = Pr_{S_m} \cup Reduc_{rank}$
- 4: **end for**
- 5: **return** Pr_{S_m}

We develop our cybersickness reduction library using GingerVR [31], an open-source software package for the Unity game engine that includes a range of cybersickness reduction

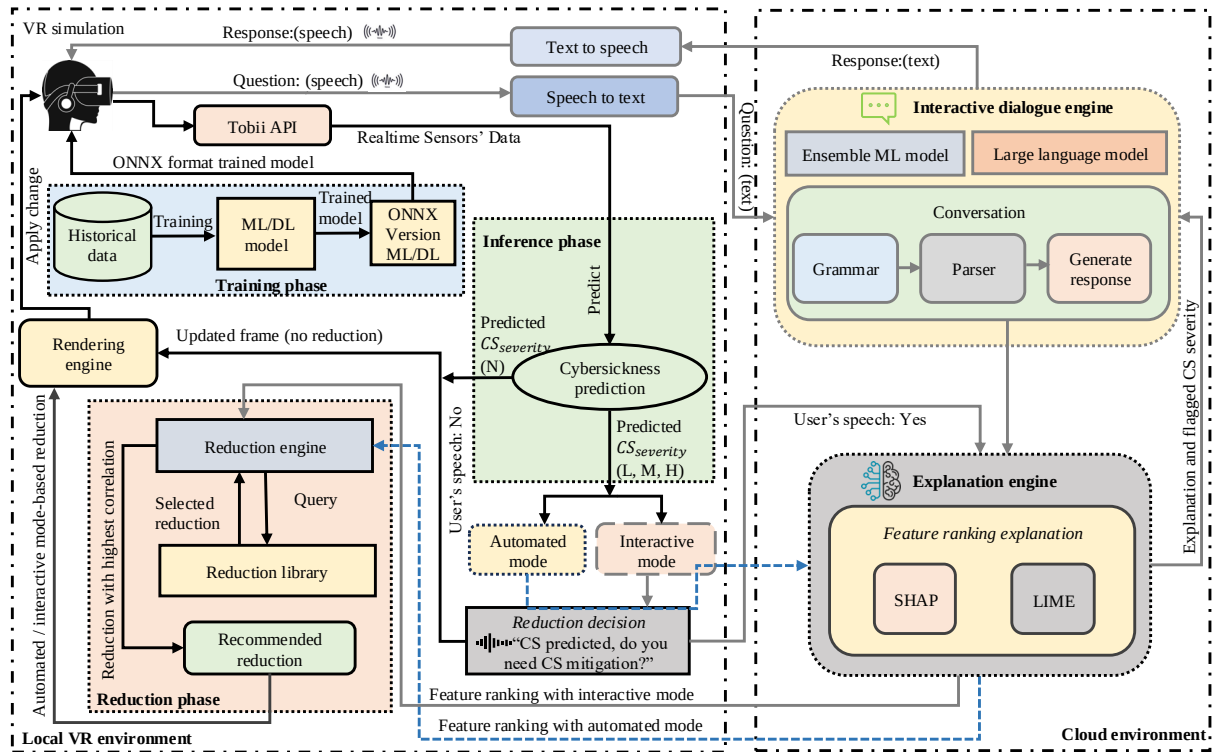


FIGURE 3: Proposed RelaxVR framework for cybersickness (CS) prediction, explanation, conversation, and reduction with two user-specified modes: 1) *automated mode* denoted with a blue dotted line that triggers XAI-guide automatic cybersickness reduction without VR user intervention during immersion and 2) *interactive mode* denoted with a gray solid line in which an interactive dialogue system using LLM along with XAI that enables VR users to interact with cybersickness detection models via voice commands to understand the reasoning behind cybersickness and how to choose effective cybersickness reduction techniques during immersion. The black solid line represent components that are common in both automated and interactive modes.

techniques, as shown in Table 3. For this study, we restrict ourselves to three commonly used reduction techniques: dynamic field-of-view (FOV), dynamic Gaussian blurring, and foveated depth-of-field (DOF) blur to build the cybersickness reduction library. It is important to note that though we use these three cybersickness reduction techniques, other available visual cybersickness reduction techniques (e.g., vision lock, rest frame, etc.) can be integrated into our cybersickness reduction library. Dynamic FOV technique uses a virtual filter to block peripheral vision areas, adjusting its center based on the user's gaze position [80]. In contrast, dynamic Gaussian blurring reduces cybersickness by decreasing the amount of optical flow accessible to the user [82]. Furthermore, a novel dynamic vignette effect manipulates FOV based on speed and gaze direction in different axes. Foveated DOF blur applies concepts from foveated imaging and depth-of-field rendering to create a seamless, artifact-free scene, which helps minimize discomfort [30]. It is important to note that each cybersickness reduction technique has advantages and limitations, as cybersickness can vary significantly between users, making a fixed reduction technique unsuitable for all [37]. For example, if the blur intensity does not align with the necessary depth level, the technique may fail to alleviate discomfort and could even disrupt the immersive

experience. Research has shown that in foveated DOF blur, users often concentrate their gaze at the center, which can inadvertently increase cybersickness. Therefore, rather than applying mitigation techniques randomly, it is essential to identify the specific features contributing to cybersickness in real-time. Our XAI-guided cybersickness reduction approach addresses this by dynamically selecting the most effective technique from the reduction library based on the feature importance scores that indicate which factors are causing cybersickness, enabling effective mitigation while preserving VR locomotion and 3D object manipulation capabilities.

We correlate the feature's importance score with the relevant reduction strategies to identify the most appropriate cybersickness reduction technique, as detailed in Algorithm 1. This algorithm takes in a ranked list of feature importance scores and various cybersickness reduction techniques as input and then returns the most suitable reduction technique. For example, suppose the explanation of predicted cybersickness indicates that the user's pupil diameter and pupil positions are the most dominant features of cybersickness in the cybersickness prediction model. In that case, the algorithm will select a reduction technique associated with these features, as detailed in Table 3. Therefore, the algorithm takes in both the list of cybersickness reduction techniques and ranked

feature importance scores and returns the probability of the most appropriate cybersickness reduction technique. For each ranked feature, the algorithm uses the list of reduction techniques CS_{reduc} and the function $CSReductionMatch()$ to calculate the probability of the most suitable reduction technique Pr_{sm} (Lines 1-2). This function computes the probability range ($Reduc_{rank}$) from 0 to 1 of the appropriate reduction technique by analyzing the absolute importance scores of each feature and their alignment with the list of cybersickness reduction techniques. Each feature's importance score contributes to ranking the reduction techniques, ensuring the most effective cybersickness reduction technique is mapped. Finally, the algorithm returns the cybersickness reduction technique with the highest probability (Line 5), which is then sent to the rendering engine.

Table 3 shows the feature and associated reduction technique to facilitate effective cybersickness reduction. We identified and designed feature correlation pairs with reduction strategies via an in-depth literature survey. For instance, if features related to the right and left eye origins are identified as the most dominant for causing cybersickness, then dynamic Gaussian blurring will be our cybersickness reduction technique. This technique minimizes conflicting visual stimuli that may disrupt the vestibular system by blurring peripheral or fast-moving elements seen by each eye [82], [100], [101]. Thus, this method effectively reduces visual discomfort and creates a more cohesive viewing experience within the virtual environment. On the other hand, if gaze-related features have the highest importance scores for causing cybersickness, then the dynamic Field of View (FOV) is our preferred cybersickness reduction technique. Prior studies have shown that adjusting the FOV dynamically based on where users are looking can significantly reduce cybersickness by addressing mismatches between visual input and inner-ear sensations, which are key causes of cybersickness [25], [80], [102], [103]. By aligning displayed content with the user's gaze direction, dynamic FOV helps reduce motion and depth discrepancies, both of which can lead to discomfort. However, if right and left pupil diameter features are the most influential in triggering cybersickness, then our chosen reduction technique is foveated Depth of Field (DOF) blur. This method selectively blurs peripheral vision based on pupil depth and gaze direction [30], [104], [105], reducing sensory conflicts and perceived motion flow during virtual movement. By gradually increasing blur intensity toward the periphery, this technique also minimizes processing load while reducing potential visual conflicts that could induce cybersickness. It is important to note that the feature importance scores are calculated based on local explanations to rank the features. For instance, Figure 4 shows an example of a local explanation for predicting cybersickness, where gaze features have high absolute importance scores. As a result, a dynamic FOV is activated to reduce cybersickness. Likewise, for each cybersickness reduction query, the absolute feature importance will be calculated and matched with the reduction technique pattern using the highest feature importance scores. This effectively reduces

TABLE 3: Cybersickness reduction library with ranked features for reduction technique chosen.

Reduction Technique	Features matching list
Dynamic Gaussian Blur [78], [82], [100], [101]	Normal Right Eye Origin (x, y, z) Normal Left Eye Origin (x, y, z) Right Pupil Position in Sensor (x, y, z) Left Pupil Diameter Left Pupil Position in Sensor (x, y, z) Right Pupil Diameter Gaze Origin World Space (x, y, z) Gaze Origin Local Space (x, y, z) Eye Openness (Left, Right)
Dynamic FOV [80], [81], [102], [103]	Gaze Origin World Space (x, y, z) Gaze Origin Local Space (x, y, z) Normal Right Eye Origin (x, y, z) Normal Left Eye Origin (x, y, z) Right Pupil Diameter Left Pupil Diameter Right Pupil Position in Sensor (x, y, z) Eye Openness (Left, Right)
Foveated DoF Blur [30], [104], [105]	Left Pupil Diameter Right Pupil Diameter Right Pupil Position in Sensor (x, y, z) Left Pupil Position in Sensor (x, y, z) Gaze Origin World Space (x, y, z) Gaze Origin Local Space (x, y, z) Gaze Direction World Space (x, y, z) Gaze Direction Local Space (x, y, z) Normal Right Eye Origin (x, y, z) Normal Left Eye Origin (x, y, z)

cybersickness for VR users while minimally impacting their user experience.

B. CLOUD ENVIRONMENT

This section describes the cloud environment, which consists of three major components: the real-time cybersickness prediction method, the explanation engine, and the LLM-guided interactive dialogue engine. The explanation engine measures the feature importance scores, ranks them, and forwards them to the cybersickness reduction engine. At the same time, the LLM-guided interactive dialogue engine enables VR users to interact with cybersickness prediction decisions through natural language conversations and helps them choose an effective cybersickness reduction technique. The following subsections discuss the details of these components.

1) Interactive dialogue engine

The interactive dialogue engine that enables *interactive mode* is based on a natural-language interface for engaging in conversations, making the cybersickness prediction model explainable and interactable. The interactive dialogue engine performs the following steps to accomplish the tasks. First, it takes user input from the predicted cybersickness severity level and a user-provided input (i.e., “What is the recommended reduction technique?”, “Why I feel cybersickness?”, etc.) via voice command. Then, the pre-trained fine-tuned large language model (LLM) is used to translate user utterances into parses. We train a pre-trained LLM on the developed *MazeSick* dataset to enable the LLM to learn the specific patterns of language specific to its intended purpose (e.g., cybersickness-related conversation generation). Next, the interactive dialogue engine performs the following steps: 1) the system constructs a grammar for the user-provided input and cybersickness model, which defines the set of

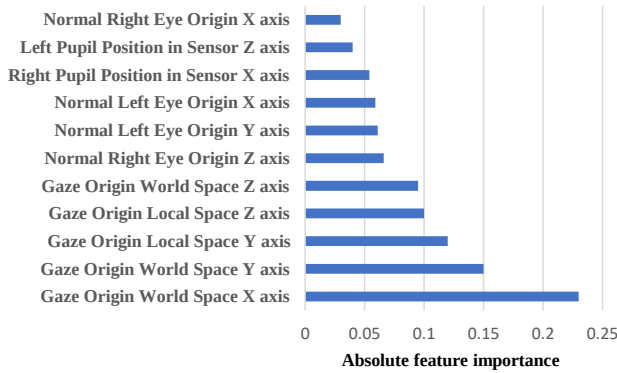


FIGURE 4: Feature importance score for single prediction

acceptable parses. Then, 2) a fine-tuned LLM consisting of utterance and parse is used to translate user utterances into parses. Finally, 3) the system responds conversationally to users by composing the results of the executed parse into a response that provides context for the results and recommendations for an effective mitigation technique. More details are as follows.

Grammar construction: The interactive dialogue engine uses grammar to represent user intentions in a structured form. This grammar consists of production rules that include the operations the system can run, the acceptable arguments for each operation, and the relations between operations. However, one complication is that VR simulation data have different feature names, making it hard to define a shared grammar between them. Thus, production rules are used to parse user utterances and generate system responses that conform to the language constraints [106]. It also ensures that the generated responses are coherent, contextually relevant, and aligned with the expected patterns of human conversation [107]. Our work updates the grammar based on the feature names in new data from the streaming VR simulation. The interactive dialogue engine also supports context in the dialogues. To execute this operation, the interactive dialogue engine relies on a set of operations in the grammar that determine the context for user utterances [107]. Namely, there are two operations: *previous filter* and *previous operation*. A previous filter extracts only the most pertinent parts of the previous conversation, such as recent interactions and user intents (i.e., “*Why I feel cybersickness?*”). This filtered information is then used to inform the generation of the current response (e.g., “*What is the recommended reduction technique?*”), ensuring that the dialogue engine focuses on the most relevant context [107]. On the other hand, a previous operation is a processing step applied to the previous dialogue history (i.e., “*What is the recommended mitigation?*”) before it is used to generate the current response (e.g., “*How to reduce cybersickness?*”) [107]. It helps refine the input context, making it more informative and conducive to generating coherent and relevant responses. Both operations look back in the conversation to determine the last filter and last

operation. These operations are also recursive, meaning that a previous filter will be recursively called to resolve the entire stack of filters if the last filter is a previous filter operation. Using these dynamics, users can generate complex parses in multiple steps, which supports natural conversations and allows much more trustworthy behavior of the generated conversation.

Parser generation: We need to parse user utterances into the grammar to provide accurate conversation. Therefore, we fine-tuned LLMs by generating a training dataset of (utterances and parse) pairs that translate utterances into grammar sequence-by-sequence, similar to the work [107]. The reason is that LLMs have been trained on large amounts of text data and have solid backgrounds in language understanding. Therefore, we use two pre-trained LLMs, namely, few-shot GPT-J [108] and fine-tuned T5 [109] for parsing user utterances into the grammar. The few-shot approach does not require fine-tuning, is quicker to set up, and is more accessible for VR users to get started with the system. T5, with fine-tuned performance, on the other hand, provides a better user experience. We incorporate a technique to select the most relevant prompts for the user to identify the closest utterances using the cosine distance [110]. Then, we evaluate the number of prompts by selecting one prompt per template included in the LLM’s context window. This work uses the GPT-J 6B, GPT-Neo 2.7B, and GPT-Neo 1.3B models from the all-mpnet-base-v2 sentence transformer model [111] for performing the word embedding.

Response generation: A known problem with parsing utterances is that their generation process may generate parses outside the grammar, resulting in the system failing to run and providing a poor user experience (e.g., misleading in effective cybersickness reduction technique selection). To assure that the generations would be grammatically correct, we confine the guided decoding [110] to take place within the grammar. After successful parsing execution, the interactive dialogue engine generates natural language responses by filling in templates associated with each operation and joining them into a final response. This approach ensures trustworthy and sufficient diversity in responses, making them suitable for effective user decision-making (e.g., selecting suitable reduction techniques).

2) Explanation Engine

This section discusses how the explanation engine automatically chooses the most accurate explanations for providing cybersickness explanations with feature rankings for individual samples. To calculate the feature importance score, the explanation engine uses two post-hoc explanation methods, namely, LIME [112] and (SHAP) [113]). Unfortunately, the explanation methods implemented as XAI tools often suffer from the *disagreement problem* [114]. A disagreement problem occurs when two explanation methods/XAI tools do not agree on the feature ranking. To alleviate this, the explanation engine provides an automatic selection of the most faithful explanation to enhance the trustworthiness of the explanation. In

this context, faithfulness is computed by perturbing the most dominant features in cybersickness severity prediction and evaluating how much the prediction changes. The evaluation is measured by computing the fudge score [107], which is the mean absolute difference between the model's prediction on the original input and a fudged version of features. Finally, the explanation with the highest fidelity value is chosen to explain the predicted cybersickness and forwarded to the cybersickness reduction engine to apply effective reduction techniques for reducing cybersickness.

VI. EXPERIMENTAL SETUP

This section explains the experimental setup and performance metrics we used to validate our proposed RelaxVR framework. We used Scikit-Learn [95] and PyTorch 2.3 [115] for training and evaluating our ensemble ML and DL models. We used the VR maze simulation to validate our proposed RelaxVR framework. Note that we used the same VR maze simulation that we used for the dataset collection task. Local RelaxVR components were integrated with the VR maze simulation to interact with users. The Cloud environment was developed using flask web framework [116] to easily connect to the local environment using HTTP requests and was deployed on an Intel Core i9 CPU with 128 GB of memory and an Nvidia GeForce RTX 3090 GPU. We used the same CPU to train the ML and DL-based cybersickness detection models. Finally, the VR simulation was displayed using an HTC Vive Pro Eye HMD with a resolution of 2880×1600 pixels per eye and a refresh rate of 90Hz.

Performance Metrics: The performance of the ensemble ML model and DL-based Transformer model in cybersickness severity classification can be quantified using standard quality metrics such as accuracy, precision, recall, and F-1 score [20]. Similarly, standard loss functions such as root-mean-squared error (RMSE), mean absolute error (MAE), and R^2 score are used to evaluate the ensemble ML and DL regression model's performance [20].

VII. TECHNICAL EVALUATION RESULTS

This section presents the technical evaluation results of the RelaxVR framework regarding cybersickness detection, explanation, and reduction.

A. CYBERSICKNESS PREDICTION MODELS PERFORMANCE COMPARISON

Before deploying the model in a VR headset, we present the trained ensemble ML and Transformer model development results with their statistics, such as training and inference time. Table 4 summarizes the accuracy, precision, recall, and F-1 scores of cybersickness severity classification using ensemble ML and DL-based Transformer models. The cybersickness classification using the ensemble ML and Transformer model resulted in an accuracy of 97%, and 95%. It is observed that the overall performance of the ensemble ML model is slightly better than the GTransformer model in terms of precision, recall, and f1-score for cybersickness

classification. Furthermore, the cybersickness regression using the ensemble ML and Transformer models results in R^2 , root mean square error (RMSE), and mean absolute error (MAE) values are 0.93, 0.12, and 0.09, for the ensemble model and 0.90, 0.16, and 0.11 for the Transformer model, respectively. Indeed, our ensemble ML model performs very well in terms of cybersickness severity classification and regression tasks. This is because models built on the ensemble technique are known for significantly improving the model performance [117]. Furthermore, each model's total training time and inference time are shown in Table 4. It is observed that the training time is significantly lower for the ensemble ML model than for the Transformer model. Furthermore, the ensemble ML model's inference times are lower than the Transformer model in the cybersickness prediction task.

B. CYBERSICKNESS PREDICTION MODEL DEPLOYMENT AND INFERENCE PHASE IN THE RELAXVR

After training and validation, the cybersickness prediction model is converted to ONNX format [98] and deployed on an HTC Vive VR headset to conduct the experiment. We used the trained ensemble ML model since the ensemble ML model performs the best in terms of cybersickness prediction tasks and requires less time in inference, as shown in Table 4. The total size of the deployed ensemble ML model on the HTC Vive Pro headset is 4.89 MB. In the VR simulation, the deployed ensemble ML model requires only 0.00018 seconds to predict the cybersickness severity level (e.g., none, low, medium, and high) per frame in real-time.

C. XAI-BASED LANGUAGE UNDERSTANDING

This section demonstrates how well RelaxVR accurately generates conversations using the LLMs and explanation engines. These explanation-based conversations show the effectiveness of RelaxVR for understanding the cybersickness model through a dialogue engine. Furthermore, we evaluate the language understanding capabilities of the cybersickness model on the streaming VR simulation data by calculating exact match parsing accuracy [118].

1) Generated Parsing Accuracy

To demonstrate the accuracy of the system's language understanding capabilities and show it effectively understands users in dialogues, we evaluate the fine-tuned T5 and GPT-J models on the developed *MazeSick* dataset using the *exact match parsing accuracy* metric inspired by the work [107], as shown in Table 5. This metric determines whether the parse exactly matches the ground truth pairs (i.e., (utterance, parse)) in the dataset. Furthermore, it ensures that while parsing the utterances, their generated parses are constrained and inside the grammar, providing better conversation and enhancing user experiences. To do this, we split the dataset into independent and identically distributed (IID), compositional, and overall splits. IID split contains (utterance, parse) pairs where the parse's operations and structure are in the training data. The compositional split consists of the remaining parses that

TABLE 4: Mean accuracy, precision, recall, and F1-score of cybersickness classification and the model's total training time (MTTT) in minutes, along with the model's total inference (Inf) in seconds for ensemble ML and Transformer models.

Models	Accuracy (%)	Precision (%)			Recall (%)			F1-score (%)			MTT (min)	Inf (Sec)
		Low	Medium	High	Low	Medium	High	Low	Medium	High		
Transformer	94	91	92	95	93	90	96	96	95	96	321	2.93
Ensemble ML	96	96	90	88	92	90	90	91	89	88	179	1.23

TABLE 5: Exact match parsing accuracy (%) for the *MazeSick* dataset, on the IID, compositional splits, and overall.

LLM model	Cybersickness		
	IID%	Comp.%	Overall%
Fine-tuned T5	85	89	88
GPT-J 6B	68	70	75

are not in the training data. The reason is that LLMs struggle compositionally for parser generation. Table 5 shows that the T5 small model performs better than the GPT-J 6B model for IID, compositional, and overall splits. For instance, the exact match parsing accuracy using the fine-tuned T5 models for compositional splits exhibits 89% accuracy, which is almost $1.25\times$ higher than that of GPT-J 6B models. Likewise, exact match parsing accuracy using fine-tuned T5 models has 85% accuracy with overall splits, almost $1.18\times$ higher than that of GPT-J 6B models. These results indicate that fine-tuned T5 small performs strongly on all the split data and can even parse complex compositional phrases in the conversation. Thus, RelaxVR can understand user utterances with high accuracy while generating the conversation.

2) Conversation Generation and Understanding

The overall conversation between the VR user with the cybersickness model for understanding the reasoning behind the flagged cybersickness and how to choose effective cybersickness reduction techniques is illustrated with an example in Figure 5. To ensure our conversation provides sufficient coverage for user queries with a model to understand these queries, we use the XAI question bank to support the grammar context, similar to [119]. Figure 5 shows that the VR user asks queries through a voice command (i.e., "Why I feel cybersickness?") to know the reasoning behind the flagged cybersickness (e.g., high cybersickness class), and the RelaxVR provides the reasoning of the flagged cybersickness. Then, the user asks for the recommended reduction technique (i.e., "What is the recommended reduction technique?") to reduce his induced discomfort. Then, RelaxVR recommends using dynamic Gaussian blurring from the mitigation library to reduce user cybersickness via mapping the feature ranking for the effective cybersickness reduction chosen. After feeling better, the user asks to stop the recommended reduction, and RelaxVR deactivates the applied reduction technique.

D. SYSTEM BENCHMARKS

To measure the end-to-end latency of the RelaxVR system for cybersickness prediction, explanation, conversation, and

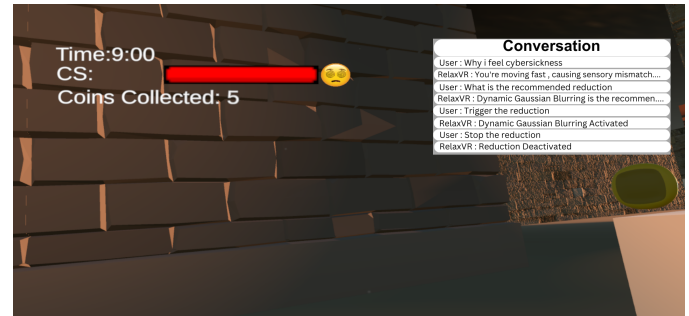


FIGURE 5: Conversation with cybersickness detection model for understanding the causing of cybersickness and recommendation for the effective cybersickness reduction through a dialogue engine.

reduction with two user-specified modes: automated and interactive mode, we measure the average time from when the cybersickness is predicted to when the recommended reduction technique is rendered during immersion for these two modes. In our implementation, we take deliberate steps to minimize latency from cloud-based processing by deploying all relevant backend services (e.g., interactive dialogue engine powered by LLM and explanation engine) on an edge cloud server co-located within the same local network as the VR headset. This approach significantly reduces communication overhead and ensures rapid data exchange between the VR and inference servers. Thus, in the automated mode, the system autonomously triggers and renders the XAI-guided proper reduction technique without VR user intervention during immersion. This mode is optimized for speed and achieves an average latency of approximately 25 milliseconds (ms), enabling near-instantaneous response within immersive VR environments. In contrast, the interactive mode involves a voice-based dialogue system using LLM along with XAI, allowing users to engage with the system through natural language to inquire about the cause of cybersickness, receive explanations, and select a preferred reduction technique. As a result, the interactive mode incurs a higher average latency of approximately 290 ms since it includes additional steps for processing user queries, generating explanations, and applying the chosen reduction technique. This latency remains within thresholds generally considered acceptable for real-time interaction in VR systems [120].

VIII. USER STUDY DESIGN ON CYBERSICKNESS REDUCTION AND EVALUATION

This section presents the detailed user study design and results of the user study evaluation to demonstrate the effective-

TABLE 6: Demographic information of the participants. The key for frequency is rarely (1 time), occasionally (a few times), and frequently (more than 2 times/month).

Demographics	# of participants (Total 26 participants)
Tye of participants	American (28%); Asian (58%); Black (14%).
Gender	10 female;16 male.
Age	Mean: 22.5; Standard deviation (SD): 4.17 years.
Frequency of VR experience	12 never used; 8 rarely; 3 occasionally; 1 frequently.

tiveness of our proposed RelaxVR framework.

A. USER STUDY DESIGN ON CYBERSICKNESS REDUCTION

To show the effectiveness of the proposed RelaxVR framework, we conducted a user study, and the details are described in the following subsections. It is important to note that we used the same apparatus (e.g., HTC-Vive Pro Eye) and VR environment (e.g., VR maze simulation) that we used for the data collection study. We already discussed these in Section III-A.

1) Participants

To evaluate the proposed RelaxVR framework, we recruited 26 participants. Although we did not conduct a formal power analysis before initiating the user study, our chosen sample size is well-aligned with prior research in cybersickness reduction [22], [26], [29], [30], [30], [77]. Prior studies have utilized participant numbers ranging from 17 to 34 to validate their proposed cybersickness reduction strategies. For example, one notable study [22] employed 25 participants to evaluate a gaze-contingent image deformation method for cybersickness reduction, while another work [23] involved 17 participants to examine an alternative cybersickness reduction approach via visual and kinematic acceleration control techniques. Furthermore, another work [29] recruited 18 participants to evaluate their proposed cybersickness reduction approach by applying vignetting techniques, and the participants experienced three conditions (e.g., Control condition without vignetting, velocity-based vignetting technique, and acceleration-based vignetting technique) during the experiments. Moreover, the authors in [30] recruited 18 participants to evaluate their proposed cybersickness reduction technique, incorporating spatial blur in VR systems to reduce induced cybersickness in virtual environments. These examples illustrate that our sample size aligns with established practices in the field. It is also worth noting that none of the above prior works conducted a power or effect size analysis before participant recruitment. Therefore, while our study did not include prior power analysis, our sample size is consistent with the norms in cybersickness reduction research and suitable for evaluating the proposed interventions. Our recruited twenty-six volunteer participants participated in the study, in which ten also participated in the *Mazesick* dataset development study. The demographic information (e.g., age, gender, racial profile, and prior VR experience) of participants is shown in Table 6. It is important to note that due to user availability and

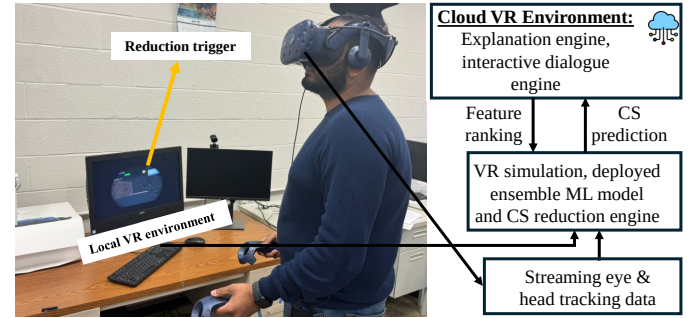


FIGURE 6: Participant immersed in the proposed RelaxVR framework

other logistical constraints, we did not recruit the same participants or the same number of participants for the RelaxVR-guided cybersickness reduction study as in the *Mazesick* dataset development study. While some participants from the dataset study took part in the cybersickness study, not all were available due to scheduling conflicts, personal commitments, or other unforeseen factors. All participants volunteered for the user study, provided written informed consent, and received no reward. No individuals reported vestibular dysfunction or the usage of anti-motion sickness medication.

2) Hypotheses

This study aims to investigate and understand strategies for effectively reducing cybersickness. Thus, to reduce cybersickness, the following two **hypotheses (H)** have been developed to serve as the foundation for an experiment, enabling the evaluation of the proposed XAI-guided cybersickness reduction method compared to conventional static/fixed cybersickness reduction techniques.

- H1: Dynamically selecting the cybersickness reduction technique using RelaxVR reduces cybersickness compared to the static cybersickness reduction technique.
- H2: Participants will have a better UIX and higher presence in the virtual environment using RelaxVR for cybersickness reduction compared to other techniques

3) Study Task and Procedure

Our user study was approved by the Institutional Review Board (IRB) at the authors' university. During the user study, the participants read the informed consent form and filled out the IRB-approved demographic information, such as age, gender, initial SSQ, disability questions, and whether they had prior immersive VR experience. Then, we gave them an overview of the user study. They were also told that they could end the study at any moment if they became too uncomfortable. Afterward, they entered the VR simulation study and fitted with the HTC Vive Pro Eye headset and controllers. The details are described below.

Warm-up Phase: Before the experiment began, we were given a concise overview of the procedures and informed participants about the study requirements. Likewise, in the *Mazesick* dataset collection user study, participants were in-

structed to verbally report their level of discomfort using an FMS scale from 0 to 10 each time a pre-recorded voice prompt played at 30-second intervals during the VR simulations. A score of 0 indicated no noticeable change in comfort level compared to the resting baseline (no sickness), while a score of 10 represented severe discomfort relative to resting conditions. Additionally, at the start of each trial, participants completed a simple practice session to familiarize themselves with the controllers and the various locomotion techniques. Likewise, in the Mazesick dataset collection user study, the participants navigated the maze using the smooth locomotion technique.

Experiment Phase: After completing the trial/tutorial session, the VR maze simulation was played and lasted 10 minutes long unless the participants decided to quit earlier. It is important to note that, likewise, in the data collection user study, the eye and head positions were carefully calibrated before initiating the simulation. Upon satisfactory calibration, participants were immersed in a VR trial session. After completing the VR simulations with and without the proposed XAI-guided cybersickness reduction techniques, participants were asked to fill out an SSQ questionnaire [39] and then take a break. Participants were allowed to rest for as long as they needed until they felt ready to continue with the next session without experiencing any discomfort. The experiment concluded once participants had completed all methods for reducing cybersickness, including the baseline condition where no mitigation was applied. To evaluate the effectiveness of our proposed RelaxVR framework and compare it with previously discussed techniques, the experiment was conducted under the following four conditions:

- **Baseline condition:** Initially, the simulation ran without applying any cybersickness reduction techniques.
- **Static reduction strategies:** Next, the cybersickness reduction techniques were applied under static conditions. In this condition, participants experienced all three reduction techniques, such as dynamic FOV reduction strategy, dynamic Gaussian blur to realize the blur effects, and foveated DOF, which incorporates DoF blur and foveation effects to provide an artifact-free scene separately one after the other within the same session, but in a randomized order. In this condition, each reduction technique was applied in a fixed, non-adaptive manner to assess its effect on cybersickness.
- **XAI-guided cybersickness reduction:** Then, as explained in Section V-A2, the XAI-guided cybersickness reduction technique was applied using the *automatic mode* from the RelaxVR, in which the system recommended a suitable cybersickness reduction technique from the reduction library.
- **XAI and LLM-interactive cybersickness reduction:** Finally, as explained in Sections V-A2 and V-B1, the user engaged with an LLM-based dialogue engine and asked to trigger the XAI-recommended reduction technique or chose user preference reduction technique using

the *interactive mode* from RelaxVR.

All of these conditions (e.g., baseline, static reduction strategies, XAI-guided cybersickness reduction, and XAI and LLM-interactive cybersickness) were conducted in counter-balanced order to ensure fair exposure and eliminate systematic bias. Each of the conditions lasted for 10 minutes. However, randomization was applied only for the static reduction strategies condition, as it contained multiple reduction techniques. Randomization was chosen only for the static condition to reduce intra-condition bias, while counterbalancing ensured balanced comparisons across all study conditions. After each of these conditions, participants were asked to complete the Igroup presence questionnaire (IPQ) [59], [121] to measure the presence for each condition. Participants were free to stop the experiment at any time if they felt significantly uncomfortable. Figure 6 illustrates the first-person view of our experimental setup, showing a participant immersed in the proposed RelaxVR framework and navigate maze and using the XAI-guided DFOV cybersickness reduction technique. As reported during the post-experiment phase, this reduction technique improved participant comfort and minimally impacted user immersive experience (UIX). Moreover, Figure 7(b), Figure 7(c), and Figure 7(d) illustrate the examples of the no mitigation condition, dynamic FOV reduction, dynamic Gaussian blur, and foveated DoF effects in detecting cybersickness during a VR maze simulation.

Post-task Interview Phase: In the post-task interview phase, we conducted an in-depth, qualitative interview with participants. At first, the participant completed a post-SSQ. Furthermore, we interviewed and collected verbal feedback from participants for about 1-2 minutes about their reflections on the study, particularly their experience of cybersickness reduction and presence across different study conditions on a 5-point Likert scale while navigating the VR maze and collecting the coins. We asked questions such as *the RelaxVR framework helped reduce cybersickness and complete the tasks, how comfortable do you feel using the RelaxVR framework than other reduction techniques throughout your VR session?*, etc., and about their experiences doing the tasks, beginning with Likert-scale questionnaires/statements to avoid priming them. Furthermore, we interviewed participants for about 3-4 minutes about their reflections on the study and collected qualitative feedback, particularly their experience with the RelaxVR framework with other conventional techniques in cybersickness reduction. Moreover, before participants departed, we emphasized that they could contact us if they experienced significant discomfort or any adverse effects resulting from the study. To date, no participants have reported any concerns.

B. CYBERSICKNESS REDUCTION USER STUDY EVALUATION

This section compares the effects of the RelaxVR framework with the different methods (e.g., with only XAI-guided and other static/fixed reduction techniques) on cybersickness reduction tasks. We begin by analyzing our experimental data to study users' reactions when encountering cybersickness

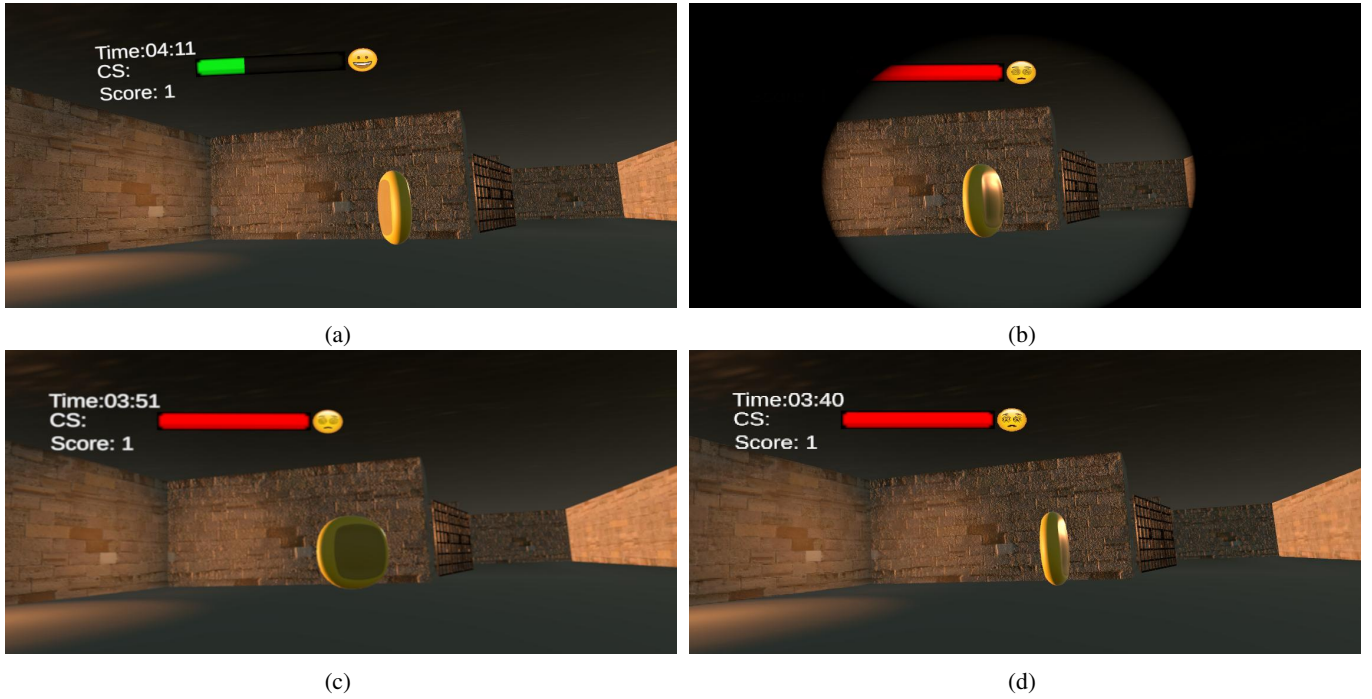


FIGURE 7: The user's perspective within the VR Maze simulation when reduction techniques are applied: (a) no mitigation applied (b) dynamic FOV reduction, (c) dynamic Gaussian blurring, and (d) foveated DOF

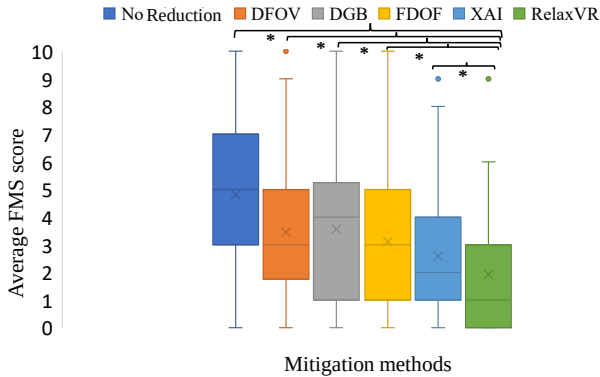


FIGURE 8: Average FMS score for different mitigation methods for 30 participants.

TABLE 7: The mean, standard deviation (SD), and 95% confidence intervals (CI) of the FMS score for each condition.

Conditions	Mean (SD)	95% CI
No mitigation	5.03 (2.64)	[4.51, 5.53]
Dynamic FOV	3.93 (2.43)	[3.59, 4.37]
Dynamic Gaussian blur	3.85 (2.38)	[3.67, 4.24]
Foveated DOF	3.67 (2.25)	[3.31, 3.98]
XAI	2.81 (2.14)	[2.58, 3.06]
RelaxVR	1.94 (1.84)	[1.61, 2.03]

using the RelaxVR framework in VR experiences. Then, we analyzed the post-task user study experimental data after encountering cybersickness in the VR experience. The details are described in the following.

TABLE 8: Wilcoxon signed-rank tests for pairwise comparison between the different mitigation conditions.

Condition	Z-value	p-value	W-value
No mitigation vs. DFOV	3.19	0.0014	1070
No mitigation vs. DGB	2.99	0.0028	1022
No mitigation vs. Foveated DOF	3.79	0.0012	860
No mitigation vs. XAI	4.95	0.0001	490
No mitigation vs. RelaxVR	6.18	0.00001	367
DFOV vs. DGB	0.3	0.763	1636
DFOV vs. Foveated DOF	1.12	0.270	1315
DFOV vs. XAI	2.14	0.026	993
DFOV vs. RelaxVR	4.16	0.0004	634
DGB vs. Foveated DOF	1.27	0.234	1391
DGB vs. XAI	1.89	0.040	865
DGB vs. RelaxVR	4.31	0.0002	730
Foveated DOF vs. XAI	1.43	0.163	1288
Foveated DOF vs. RelaxVR	3.13	0.001	940
XAI vs. RelaxVR	1.97	0.049	1017

TABLE 9: Comparison among different mitigation techniques for reducing cybersickness

Reduction techniques	Δ FMS
Dynamic FOV	58.8%
Dynamic gaussian blurring	60.1%
Foveated DoF	64.5%
XAI	73.4%
RelaxVR	84.6%

Experiment phase: The descriptive statistics (e.g., mean, standard deviation, and 95% confidence interval) of each condition's measured FMS score are shown in Table 7. The results show that the FMS score is lowest in RelaxVR conditions, ranging between 1.61 – 2.03 in the 95% confidence

intervals. In contrast, the measured FMS score is relatively higher in the XAI-guided and static cybersickness reduction conditions. To test the significance of these differences in means, we first conduct a Shapiro-Wilk test, after which we determine that the FMS score per 30 seconds of data is not normally distributed. Therefore, we conduct a non-parametric test, namely the Friedman test, on these kinds of data instead of ANOVA. We set our significance level $Alpha(\alpha) = 0.05$. The Friedman test indicates that there are significant differences in cybersickness between the conditions: the baseline, DFOV, DGB, Foveated FOV, XAI-guided, and RelaxVR ($\chi^2 = 40.45$, $p = 0.00001$). For post hoc comparisons, we conduct Wilcoxon signed-rank tests [122] for each pair, as shown in Table 8. In the Wilcoxon signed-rank tests, we calculate the W-value, p-value, and Z-value to quantify the difference between different conditions. From Table 8, we observe that all reduction techniques show a reduction of cybersickness vs. no reduction. Moreover, we find significant differences between each of the static methods vs. RelaxVR - **DFOV vs. RelaxVR** ($Z = 4.16$, $W = 634$, $p = 0.0004$), **DGB vs. RelaxVR** ($Z = 4.31$, $W = 730$, $p = 0.0002$), **Foveated DOF vs. RelaxVR** ($Z = 3.13$, $W = 940$, $p = 0.001$), and **XAI vs. RelaxVR** ($Z = 1.97$, $W = 1017$, $p = 0.049$). However, we do not find any significant difference when comparing each static mitigation technique to the other static cybersickness reduction techniques (p-values > 0.05). Figure 8(a) shows the average FMS score of the different applied reduction techniques in cybersickness reduction for different users. This box plot shows that our proposed RelaxVR method outperforms other methods in reducing cybersickness by significantly reducing users' FMS scores, which supports and accepts our **H1**. Furthermore, to evaluate different reduction techniques for cybersickness reduction, we measure the reduction of mean FMS scores (Δ FMS), which are the differences in the FMS scores between the no-effect (without reduction) condition and the best-performing condition similar to the work [30]. Table 9 compares different reduction techniques for reducing cybersickness with our RelaxVR method. Our RelaxVR method outperforms the other techniques in cybersickness reduction. For instance, the RelaxVR method obtains a reduction of mean FMS scores of 84.6%, which is $1.44\times$, $1.4\times$, $1.31\times$, and $1.15\times$ higher than dynamic FOV, dynamic Gaussian blurring, foveated DOF, and XAI-guided methods, respectively. Overall, these results support our **H1**, which shows that dynamically selecting the proper reduction technique using RelaxVR significantly reduces cybersickness compared to static reduction techniques.

To evaluate the proposed RelaxVR condition with other conditions (e.g., no reduction, static, and XAI) regarding presence, we collect presence data from IPQ presence questionnaires during the experiment phase. The questionnaire used a seven-point Likert scale to register the feedback of the user testing participants. The analysis of presence scores (i.e., IPQ) across the four study conditions reveals notable differences. The descriptive statistics (e.g., mean (M), standard deviation (SD), and 95% confidence interval) of Re-

TABLE 10: Descriptive statistics for the Presence score of different study conditions

Condition	Mean (SD)	95% Confidence Interval
No reduction	2.34 (1.82)	[1.55, 2.63]
Static	3.61 (1.76)	[3.12, 4.19]
XAI	4.09 (2.17)	[3.40, 4.73]
RelaxVR	5.29 (1.69)	[4.82, 5.87]

TABLE 11: Mann-Whitney U tests comparing presence score for different study conditions

Conditions	Z-value	P-value	U-value
No reduction vs. Static	2.95	0.0033	556
No reduction vs. XAI	3.26	0.0017	533
No reduction vs. RelaxVR	5.52	0.0001	262
Static vs. XAI	0.96	0.3489	779
Static vs. RelaxVR	3.71	0.003	466
XAI vs. RelaxVR	2.33	0.013	601

laxVR condition and other study condition methods measured presence scores are shown in Table 10. We observe the highest presence scores in the RelaxVR condition ($M = 5.29$, $SD = 1.69$) and the lowest in the no reduction condition ($M = 2.34$, $SD = 1.82$). Furthermore, the 95% confidence interval for the means indicates that the higher presence score is in the RelaxVR condition and then XAI-guided condition without conversation. Furthermore, we also analyze the participants' presence score's normality test to examine the score distribution using the Shapiro-Wilk test and ensure that the sample data is not drawn from a normally distributed population. Therefore, we perform a post-hoc Mann-Whitney U tests [123] to find statistical significance between the different study conditions, as shown in Tables 11. We calculate the Z-value, p-value, and U-value in the Mann-Whitney U tests to quantify the difference between different study conditions of the gathered presence score. Upon examining this data, we find significant differences between RelaxVR and other study conditions (e.g., no reduction, static, and XAI) (p-values < 0.05). However, we do not find any significant difference when comparing **static vs. XAI** condition (p-values > 0.05), as shown in Table 11. Thus, this finding suggests that cybersickness significantly affects a user's sense of presence in VR while using the RelaxVR framework, which substantially provides a higher sense of presence by reducing cybersickness and supporting our **H2**.

Post-task interview Phase: We also provide additional user study results demonstrating how well the RelaxVR provides comfortability and effective cybersickness reduction compared to the other reduction methods. For this, during the user study, we asked users to rate their experiences using the RelaxVR framework on a 1 to 5 in the Likert scale [124] compared to using other reduction techniques (e.g., only XAI guided without conversation and static reduction techniques). More specifically, we asked users how much they agree with the following statements (S): 1) *The framework helped reduce cybersickness and complete the tasks (e.g., stay engaged in the VR simulation)*, 2) *Based on my experience and using*

the interface so far, the decision regarding cybersickness is reliable, 3) Using RelaxVR method enhance your overall VR experience., 4) While using the RelaxVR framework for cybersickness mitigation, you felt comfortable throughout your VR session, 5) The RelaxVR framework helped you maintain a higher presence level throughout the tasks, and 6) Would you recommend this RelaxVR method than other methods to others who use VR? We select these six specific statements to measure the developed RelaxVR system's performance because they collectively capture the essential aspects of user experience and system effectiveness, which are critical for evaluating a new technology to reduce cybersickness and enhance VR engagement. For instance, the statement (i.e., *The framework helped reduce cybersickness and complete the tasks (e.g., stay engaged in the VR simulation)*) directly evaluates the RelaxVR method's effectiveness in reducing cybersickness while maintaining user engagement in VR. Similarly, another statement (i.e., *The RelaxVR framework helped you maintain a higher presence level throughout the tasks*) assessing RelaxVR's effectiveness in reducing cybersickness while maintaining immersion. By evaluating whether RelaxVR helps sustain presence, this question measures its impact on comfort and user experience, ensuring an effective and immersive VR environment. Furthermore, statement (i.e., *Would you recommend this RelaxVR method than other methods to others who use VR?*) measures user satisfaction and endorsement, with high recommendation rates indicating strong user approval and perceived value [125]. Therefore, these six statements were selected for their comprehensive insights into effectiveness, impact, comfort, and user preference. We computed the mean and standard deviation of Likert scores for these queries, comparing RelaxVR with other methods. As shown in Figure 9, 94% of users favored RelaxVR for cybersickness reduction and engagement. In contrast, most users agreed not to use the other methods due to their ineffective user immersive experience in the VR environment. Thus, these results support **H2**, confirming that RelaxVR enhances UX and overall VR experience more effectively while maintaining a higher presence than other methods.

SSQ Results for All Study Conditions: We find a significant difference while comparing the pre-study SSQ scores and the post-study SSQ scores for all participants. For this, we first perform a Shapiro-Wilk test on these data, indicating that we cannot assume normality. Therefore, we conduct the Wilcoxon signed-rank test to quantify the difference between the pre and post-session SSQ subscores and the total scores. All cybersickness subscores except Disorientation (p-values > 0.05) indicated a significant increase post-session relative to pre-session: Nausea ($Z = -3.46$, $W = 88$, $p = 0.019$), Oculomotor ($Z = -2.95$, $W = 59$, $p = 0.003$), Disorientation ($Z = -1.65$, $W = 115$, $p = 0.078$), and Total Score ($Z = -2.98$, $W = 60$, $p = 0.0015$).

IX. DISCUSSION

This section briefly discusses the results obtained using the RelaxVR framework. Both technical evaluations and user

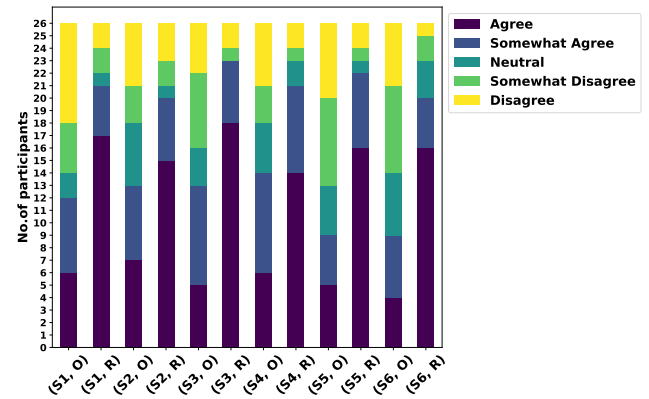


FIGURE 9: User feedback in the Likert scale between the RelaxVR framework denoted as **(R)** and without the RelaxVR framework denoted as the other methods **(O)** technique for cybersickness reduction.

studies have demonstrated that the RelaxVR method effectively reduces cybersickness and provides a higher sense of presence without compromising the user experience (UX). After engaging with the RelaxVR framework and applying XAI-driven and user-preferential reduction strategies through the dialogue engine, most participants reported lower discomfort and disorientation compared to static and XAI (without conversation) reduction, supporting our hypothesis (**H1**). Our findings indicate that XAI, with a user preference-based reduction approach, achieved an 84.6% reduction in mean FMS scores. In addition, the interactive dialogue system guided by LLM allowed VR users to select effective reduction techniques through natural language interactions. Moreover, our explanation selection method identified the most faithful explanations from multiple approaches, generating feature rankings that assisted VR users in selecting the most effective reduction strategy from the cybersickness reduction library. Finally, the findings indicate that most users preferred the RelaxVR framework over other approaches (e.g., static and only XAI-guided) (see Figure 9). Furthermore, user feedback during our study confirmed that this level of latency did not disrupt immersion or hinder interaction flow. Participants found that the interactive mode of the RelaxVR system is responsive and helpful to understand the reasoning behind cybersickness and choose effective reduction techniques to reduce cybersickness during immersion.

Since no previous work has applied XAI to reduce cybersickness, our results are not directly comparable with prior works [14], [22]–[24], [26], [28]–[30], [38], [75]–[77]. However, we compared our results regarding user comfort when they used our RelaxVR framework with static cybersickness reduction techniques. Previous works dynamically adjusted cybersickness reduction techniques without identifying the essential features behind causing cybersickness [32], [38], potentially disrupting UX and sense of presence. For instance, Kim et al. [38] applied a fixed type of reduction technique (i.e., reduce FOV) without knowing the reason behind the

cybersickness and without adjusting its magnitude to match cybersickness intensity, potentially diminishing UX. While Islam et al. [32] adaptively adjusted cybersickness reduction techniques without identifying its underlying factors, potentially disrupting immersion, worsening symptoms, and increasing cybersickness severity. The reason is that a static reduction technique, such as fixed dynamic FOV or Gaussian blurring, risks applying a one-size-fits-all solution that fails to account for real-time user experience and symptoms. On the other hand, our proposed RelaxVR method identifies the important features for causing cybersickness by the ML-based detection model by calculating the feature importance score and mapping that to an appropriate reduction technique for effective cybersickness reduction. For instance, after experiencing the dynamic FOV reduction method, one participant **P4** reported that using the static foveated DOF technique didn't work well for him because it applied the blur uniformly without considering where he was looking, making it harder to focus on collecting coins thus increased his discomfort, and potentially led him to leave the VR simulation. On the other hand, while the same participant **P4** experienced the RelaxVR recommended dynamic Gaussian blurring reduction method, adjusted the blur dynamically, keeping him comfortable without disrupting his ability to see and interact with the VR environment. This is because the dynamic Gaussian blurring technique reduces the perceived motion blur and visual discomfort caused by rapid movements in the user's peripheral vision, thereby mitigating the sensory conflicts contributing to cybersickness [82], [100]. Another **P12** reported that the XAI-guided approach (without conversation) was better than static mitigation but still lacked flexibility. Sometimes, she wished to change the reduction technique but couldn't. However, with RelaxVR, he could simply ask the system to trigger her reliable reduction technique (e.g., dynamic Gaussian blurring). After that, she found a huge difference in her comfort and presence during the VR navigation. Another **P9** reported that the interactive dialogue in RelaxVR made him feel like the system understood his needs. He could ask it to apply a specific technique, and it responded immediately. He found this was much better than the XAI-only approach, where he had no choice in what was applied. While another **P14** said, she felt significantly more presence with RelaxVR. The static reduction technique (e.g., dynamic FOV) didn't work for her because it narrowed her field of view too aggressively in some areas of the maze, making it harder to navigate and, increasing her cybersickness and breaking her immersion. She found that RelaxVR was smarter in applying the proper technique at the right time, which helped her complete the task without breaking the immersion. Thus, these results also support **H2**, in which participants had a better UX and a higher sense of presence that enhanced the overall VR experience when using the RelaxVR framework (Table 10).

Previous works has conducted different user studies and explored various cybersickness reduction techniques randomly for reducing cybersickness [14], [28]–[30], [38]. In contrast, little research has been conducted on XAI to identify the

features indicating cybersickness [19]–[21]. However, to our knowledge, there is no work on applying XAI and LLM for effective cybersickness reduction technique development. Indeed, XAI and LLM-guided interactive cybersickness reduction systems can provide structured insights, helping researchers and VR users understand the effects of visual reduction techniques, such as optimizing dynamic FOV adjustments and determining appropriate blur levels for Gaussian blurring across different VR scenarios. Thus, we believe that the RelaxVR framework offers a practical solution for developers, testers, and VR users, enabling systematic, real-time cybersickness reduction for enhanced on-device VR experiences.

X. APPLICABILITY OF RELAXVR FRAMEWORK FOR OTHER VR GAMES

The presented RelaxVR framework, which is evaluated in a VR maze-navigation environment, is highly adaptable and can be extended to a wide range of VR game genres, such as roller coaster, shooter, track/car racing, beach city exploration, waypoint navigation, sea voyage experiences, etc., with varying visual intensity, interaction styles, and motion patterns [14], [24], [25], [47], [126], [127]. For instance, in high-motion experience games such as VR roller coasters, shooters, and car racing, RelaxVR can predict cybersickness severity using the deployed ensemble ML model and then apply the XAI-guided cybersickness reduction technique (e.g., dynamic FOV) from the cybersickness reduction engine in real-time. By dynamically narrowing FOV during rapid movements, users experience a reduced sense of speed and motion intensity, which diminishes the *vection effect* [128], a primary contributor to cybersickness in VR roller coaster environments. While for slower-paced or exploration-focused games such as Beach City, waypoint navigation, or sea voyage simulations, RelaxVR can predict the severity of cybersickness. The explanation engine analyzes feature importance scores to identify which physiological or behavioral signals contribute most to the discomfort. Based on this explanation, the system selects the most suitable reduction technique from a reduction library. For instance, if increased gaze instability and head jitter features are identified as dominant contributors to predicted sickness, RelaxVR may automatically trigger dynamic Gaussian blurring, which softly reduces visual detail, specifically in peripheral vision during motion or gaze instability events. This reduces the user's sensitivity to subtle motion cues and suppresses the strength of the visual flow, helping to alleviate discomfort from mild but continuous sensory conflict. Furthermore, the integrated LLM-based dialogue system further enhances adaptability by allowing VR users to ask natural-language questions (i.e., "Why I feel cybersickness?", "What is the recommended reduction technique?", etc.), and receive explanations regarding cybersickness, and select a preferred reduction technique. These capabilities make RelaxVR a generalizable and user-centric framework that can improve comfort, immersion, and usability across a broad spectrum of VR applications while

remaining sensitive to each game type's unique interaction and motion characteristics.

XI. LIMITATION AND FUTURE WORKS

Our RelaxVR framework has some limitations. First, our user study was conducted with a single population (participants experiencing only a VR maze simulation) and had imbalanced gender representation without considering individual factors such as age, task type, or exposure duration. Since cybersickness can vary based on their unique characteristics, tasks, and VR environments [6], [77], hence it is essential to evaluate RelaxVR across different VR content, including gaming and interactive scenarios, with longer exposure times and a more diverse, gender-balanced participant pool. Moreover, this study used a single VR simulation and three visual cybersickness reduction techniques. Thus, in the future, we plan to run experiments with other VR simulation environments and add more cybersickness reduction techniques to the cybersickness reduction library. Furthermore, this study primarily measured the immediate effectiveness of RelaxVR without integrating user feedback for correlation improvement. Therefore, We plan to incorporate user feedback into the system in the future, enabling personalized cybersickness prediction and reduction through model retraining.

XII. CONCLUSION

In this work, we proposed the RelaxVR, an interactive VR cybersickness reduction framework, by leveraging XAI for cybersickness predictions, explanations, and reduction through an interactive LLM-based dialogue engine for conversation. Specifically, we constructed a cybersickness dataset, namely the MazeSick dataset, which contains multimodal time-series sensor data (e.g., eye-tracking and head-tracking) and cybersickness levels collected from 60 participants through a user study. Then, we developed an efficient cybersickness detection model using this Mazesick dataset with a proper explanation selection method to provide accurate feature rankings from the detection model. These are then mapped to appropriate cybersickness reduction techniques and recommended to users. We implemented RelaxVR on the HTC Vive Pro and conducted a mixed factorial user study, comparing the effects of different cybersickness reduction techniques with RelaxVR. Our user study revealed that the aptness of our proposed RelaxVR significantly reduces VR cybersickness with minimal impact on user immersion (UIX), and 94% of users agreed it is highly effective and easy to use for interacting with the cybersickness detection model through the dialogue engine. We believe that the RelaxVR framework will be helpful for other VR researchers, developers, and VR users for developing, testing, and using real-time cybersickness detection and mitigation methods on consumer-level VR headsets.

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